

$\det \mathcal{G}'_h = 1$. Thus the asymmetric variant of the Euler method is area preserving, even though the standard Euler method is not.

It would be valuable to have a general method for deriving volume conserving methods. It turns out that volume conservation is, itself, most readily obtained as a consequence of a more fundamental property of Hamiltonian flows, the conservation of the symplectic form.

2.3.3 The Symplectic Property

Let $m = 2N_c$. A symplectic map $\Phi : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is one that preserves the symplectic differential 2-form. The simplest way to write this is as the following algebraic condition on the Jacobian matrix of Φ :

$$\Phi'^T J \Phi' = J.$$

Recall that the Jacobian matrix is the $m \times m$ matrix of partial derivatives of the components of Φ , $(\Phi')_{ij} = \frac{\partial \Phi_i}{\partial z_j}$, while

$$J = \begin{bmatrix} \mathbf{0} & I \\ -I & \mathbf{0} \end{bmatrix}.$$

So the equation $\Phi'^T J \Phi' = J$ represents a large number ($4N_c^2$) of equations.

To explain the origin of this formula, we need a brief diversion into the world of differential forms. This language will also be useful in Chap. 4, where we consider constrained systems.

A 1-form α defined on $\mathbb{R}^m$ is a family of linear mappings from $\mathbb{R}^m$ to $\mathbb{R}$, defined for each point of $\mathbb{R}^m$. Let $\mathbf{a} : \mathbb{R}^m \rightarrow \mathbb{R}^m$, then we may define a one-form associated to this vector by $\alpha(\mathbf{x})(\xi) = \mathbf{a}(\mathbf{x})^T \xi$.

The differential of a function $g : \mathbb{R}^m \rightarrow \mathbb{R}$, denoted dg , is a family of linear mappings (one for each point in phase space) from vectors $\xi \in \mathbb{R}^m$ into the reals defined by

$$dg(\mathbf{q}, \mathbf{p})(\xi) = \nabla g(\mathbf{q}, \mathbf{p})^T \xi.$$

So, denoting the i th position coordinate by q_i , we have $dq_i(\xi) = \xi_i$; the differential is thus an example of a 1-form.

On the other hand the so called *wedge product* of 1-forms α , β , is an example of a 2-form, at any point in phase space it can be viewed as a quadratic form, i.e. a scalar valued function of two vectors which is linear in each argument. It is written $\alpha \wedge \beta$ and is defined, for vectors $\xi, \eta \in \mathbb{R}^m$ by

$$\alpha \wedge \beta(\xi, \eta) = \alpha(\xi)\beta(\eta) - \alpha(\eta)\beta(\xi).$$